NAME: _____ Geometry (FL B.E.S.T.)

4.2 Isosceles & Equilateral Triangle Theorems

The **Base Angles Theorem** states that if two sides of a triangle are congruent, the angles opposite those sides (the base angles) are congruent. Its **converse** is also true: if two angles of a triangle are congruent, the sides opposite them are congruent. An equilateral triangle is a special case where all three sides are congruent; applying the Base Angles Theorem to each pair of equal sides shows all three angles must also be congruent, and since they sum to $180°$, each equals exactly $60°$ (the Equilateral Triangle Corollary). Conversely, an equiangular triangle must be equilateral. This worksheet applies these theorems with numeric problems, algebra, coordinates, and a short proof.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Isosceles triangle, vertex angle $50°$. Find each base angle.

2. Isosceles triangle, base angles $62°$ each. Find the vertex angle.

3. In $\triangle ABC$, $\angle B \cong \angle C$. What can you conclude about $\overline{AB}$ and $\overline{AC}$?

4. Find each angle of an equilateral triangle.

5. Base angles are $4x$ and $3x + 12$. Find x and each base angle.

6. Vertex angle is x, base angles are $2x$ each. Find x and all three angles.

7. Equilateral triangle has side length 9. Find the perimeter.

8. $AB = AC$ in $\triangle ABC$. Which two angles are the base angles?

9. Isosceles triangle, vertex angle $110°$. Find each base angle.

10. An equiangular triangle is also what type of triangle by sides?

11. Base angles are $5x - 4$ and $3x + 10$. Find x, each base angle, and the vertex angle.

12. Isosceles triangle, base angles $48°$ each. Find the vertex angle.

NAME: _____ Geometry (FL B.E.S.T.)

13. An equilateral triangle has two sides represented by $4x - 3$ and $2x + 7$. Find x, the side length, and the perimeter.

14. Vertices $A(0, 6)$, $B(-8, 0)$, $C(8, 0)$. Find all three side lengths, identify the vertex angle and base angles, and find the perimeter.

15. An isosceles roof truss has each base angle $= x$ and vertex angle $= 3x - 10$. Find x, all three angles, and classify by angles.

✓ Answer Key

1. $65°$
2. $56°$
3. $\overline{AB} \cong \overline{AC}$
4. $60°$
5. $x = 12$; base angles $48°$ each
6. $x = 36$; angles $36°, 72°, 72°$
7. 27
8. $\angle B$ and $\angle C$
9. $35°$
10. Equilateral
11. $x = 7$; base angles $31°$ each; vertex $118°$
12. $84°$
13. $x = 5$; side $= 17$; perimeter $= 51$
14. $AB = AC = 10$, $BC = 16$; vertex angle at A, base angles at B and C; perimeter $= 36$
15. $x = 38$; base angles $38°$ each, vertex $104°$ — obtuse isosceles